

AWS IAM Assume Role for Cross Account Access

To Begin with the Lab

Summary of the Lab

In this lab, **AWS IAM assume role** was used to provide temporary access across two separate AWS accounts. The source account, represented by **Cloud Store Private Limited**, hosted an **Amazon S3** bucket containing high-resolution images. A role was created in the Cloud Store account with **Amazon S3 Full Access**, and the trusted AWS account ID from **Photo Magic Private Limited** was added so that users from that account could assume the role. In the Photo Magic account, an IAM user named **Magic Editor** was created and given permission to assume the cross-account role. After switching roles, the user was able to access, download, and upload objects in the Cloud Store S3 bucket temporarily, and could switch back to the original identity afterward.

Understanding Cross-Account Assume Role

Cross-account assume role is an IAM feature that lets one AWS account temporarily grant access to resources in another AWS account. It exists so organizations do not have to create and manage permanent IAM users for external teams, partners, or contractors. Instead, the source account creates a role and trusts the other AWS account to assume it. AWS STS then provides temporary credentials, which improves security, simplifies administration, and makes access easier to audit.

NOTE: The names here are for demonstration only, I will be using my own accounts for demonstration but example are sudo-names for a better understanding.

cf-shashank1 is taken as **Cloud Store Private Limited**.

iam_shashank_ac2 is taken as **Photo Magic Private Limited**.

Example:

Cloud Store stores design images in an S3 bucket named HDPictures01. Photo Magic needs to access those images for editing work. Rather than creating a permanent Cloud Store IAM user for Photo Magic, Cloud Store creates a role that Photo Magic's IAM user can assume only when needed. The editor can then temporarily access the bucket and switch back afterward.

Lab Steps

Step 1 – Create the Source S3 Bucket in the Cloud Store Account

- Sign in to the **Cloud Store Private Limited** AWS account.
- Open **Amazon S3**.

Amazon S3

Buckets

General purpose buckets | All AWS Regions | Directory buckets

General purpose buckets (225) Info

Buckets are containers for data stored in S3.

Find buckets by name

	Name	AWS Region	Creation date
☐	ab-aws-de-labs	Asia Pacific (Mumbai) ap-south-1	February 12, 2026, 00:44:57 (UTC+05:30)
☐	amazon-datawarehouse-nv	US East (N. Virginia) us-east-1	June 16, 2026, 15:49:49 (UTC+05:30)
☐	amazon-datawarehouse1df	Asia Pacific (Mumbai) ap-south-1	June 16, 2026, 15:48:52 (UTC+05:30)
☐	amazon-cdimagesdata	Asia Pacific (Mumbai) ap-south-1	November 13, 2025, 15:18:01 (UTC+05:30)
☐	amazon-s3-bdmampdata2201	Asia Pacific (Mumbai) ap-south-1	January 22, 2026, 11:32:08 (UTC+05:30)
☒	amazon-officalapps	November 12, 2025, 11:03:13 (UTC+05:30)	
☒	amazon-prk1	November 4, 2025, 18:41:42 (UTC+05:30)	
☒	amazon-s3-staticwebinfo1	December 27, 2025, 11:32:40 (UTC+05:30)	
☒	amazon-web-data1na	May 22, 2026, 20:59:53 (UTC+05:30)	
☒	amazonweb0data1	April 19, 2026, 14:37:36 (UTC+05:30)	

Account snapshot Info View dashboard

Updated daily

Storage Lens provides visibility into storage usage and activity trends.

External access summary Info

Updated daily

External access findings help you identify bucket permissions that allow public access or access from other AWS accounts.

- Click **Create bucket**.
- Enter the bucket name: updated-photos-01.

Create bucket Info

Buckets are containers for data stored in S3.

General configuration

AWS Region
US East (N. Virginia) us-east-1

Bucket type Info

☒ **General purpose**
Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.

☐ **Directory**
Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.

Bucket namespace
Choose the namespace where you want to create your bucket. [Learn more](#)

☒ **Global namespace**
By default, S3 creates general purpose buckets in the global namespace.

☐ **Account Regional namespace (recommended)**
General purpose buckets created in your account Regional namespace are unique to your account. These buckets can never be created by another AWS account.

Bucket name Info

updated-photos01

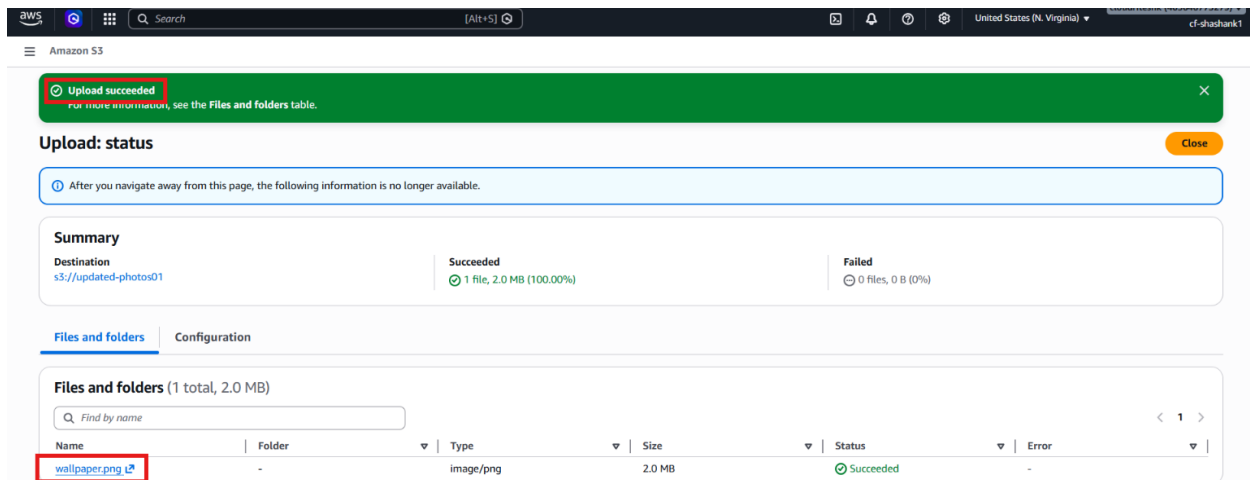
Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). [Learn more](#)

Copy settings from existing bucket - optional
Only the bucket settings in the following configuration are copied.

[Choose bucket](#)

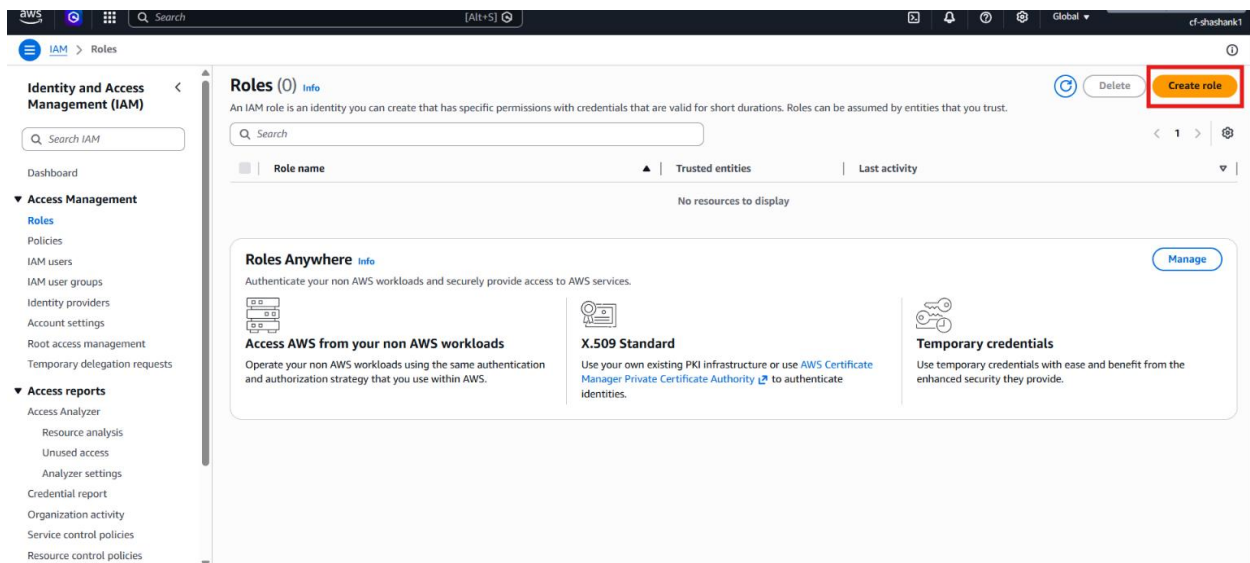
Format: s3://bucket/prefix

- Upload the image files that Photo Magic needs.
- Verify that the objects are present in the bucket.

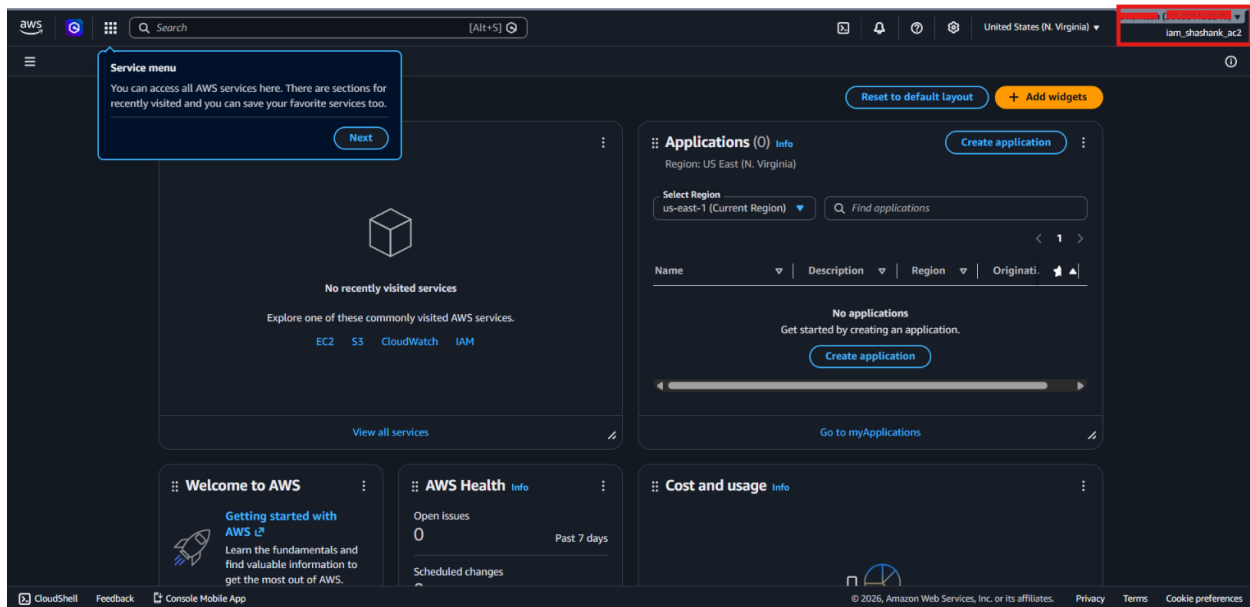


Step 2 – Create the Cross-Account Role in Cloud Store

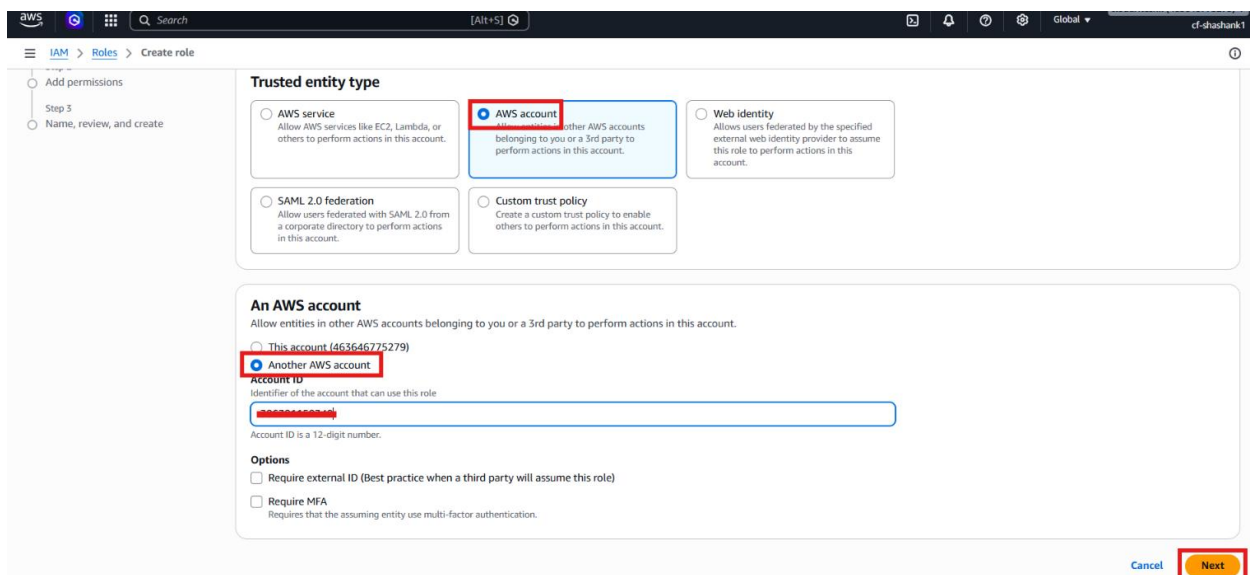
- Open IAM in the Cloud Store account.
- Click Roles → Create role.



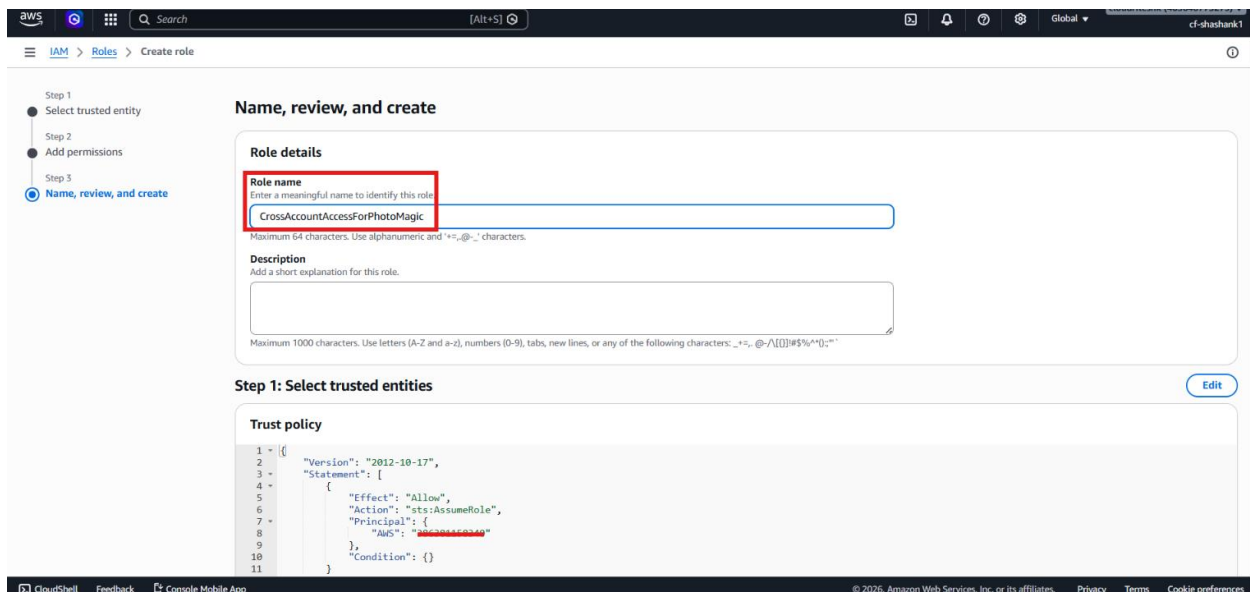
- Select **AWS account**.
- Choose **Another AWS account**.
- Enter the AWS account ID for **Photo Magic Private Limited** which can be found here for reference.



• Click Next.



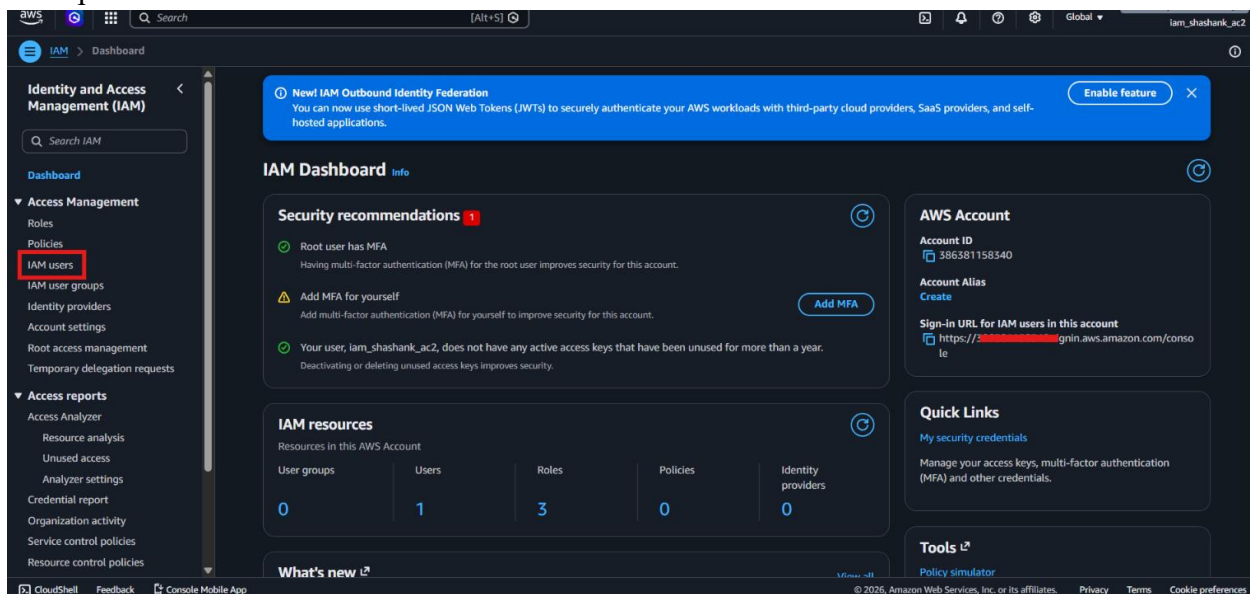
- Attach the required permission policy.
- For this lab, select **AmazonS3FullAccess**.



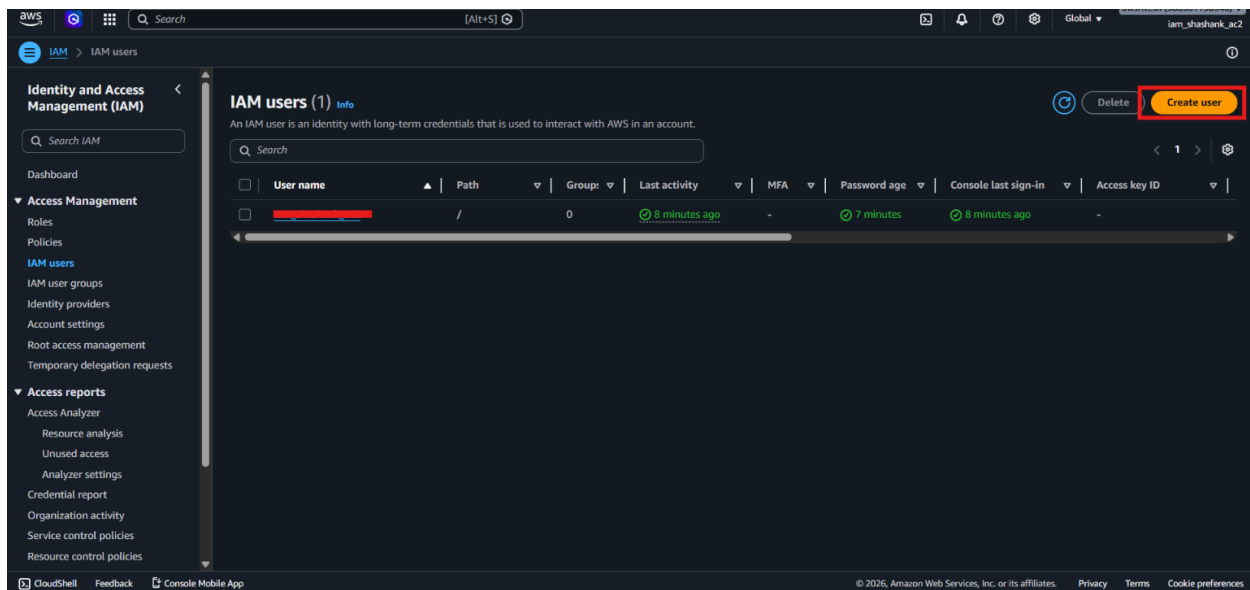
- Name the role: CrossAccountAccessForPhotoMagic
- Create the role.
- Confirm that the trust relationship allows the Photo Magic AWS account to assume the role.

Step 3 – Create the IAM User in Photo Magic

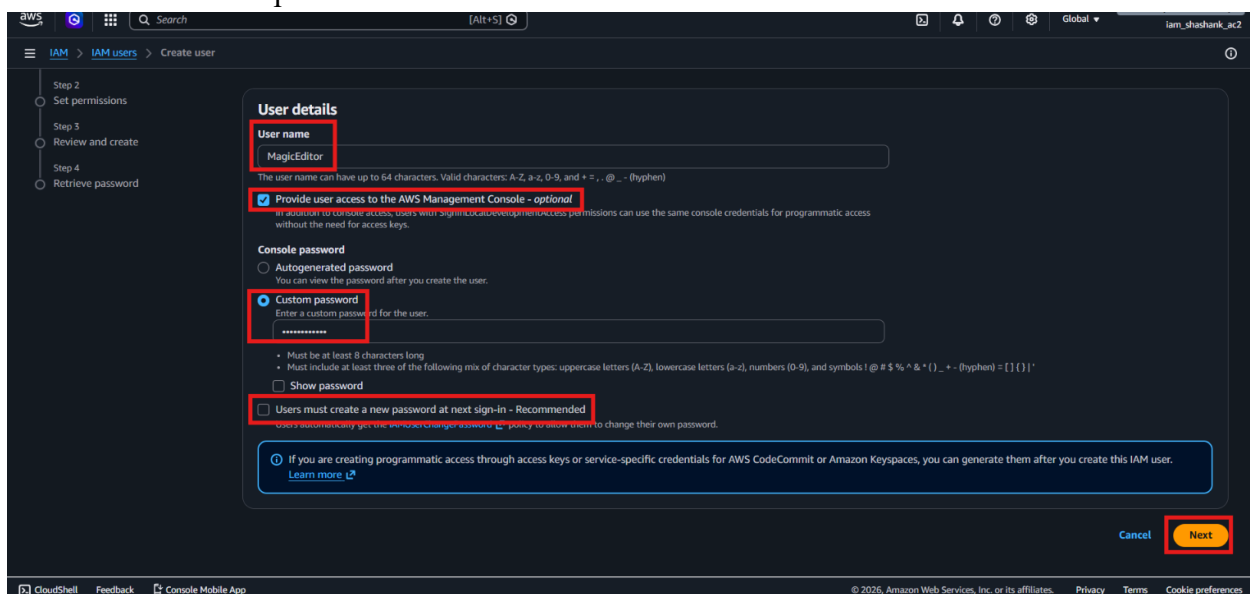
- Sign in to the **Photo Magic Private Limited** AWS account.
- Open IAM → IAM Users.



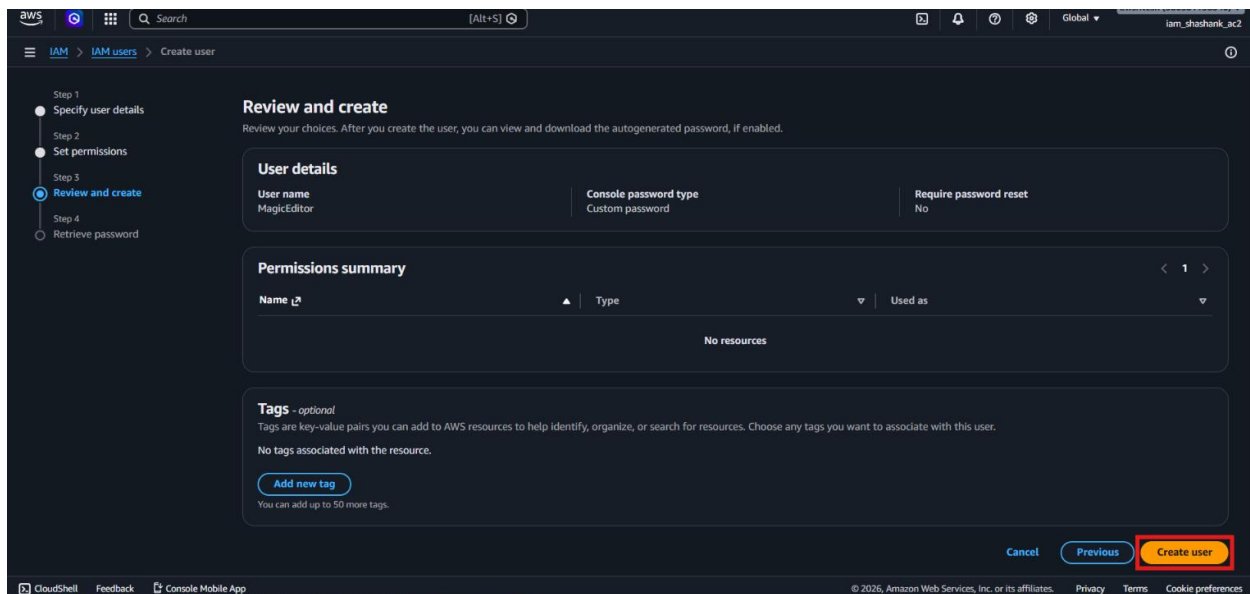
- Click **Create user**.



- Enter the user name: MagicEditor
- Enable AWS Console access.
- Create a console password.

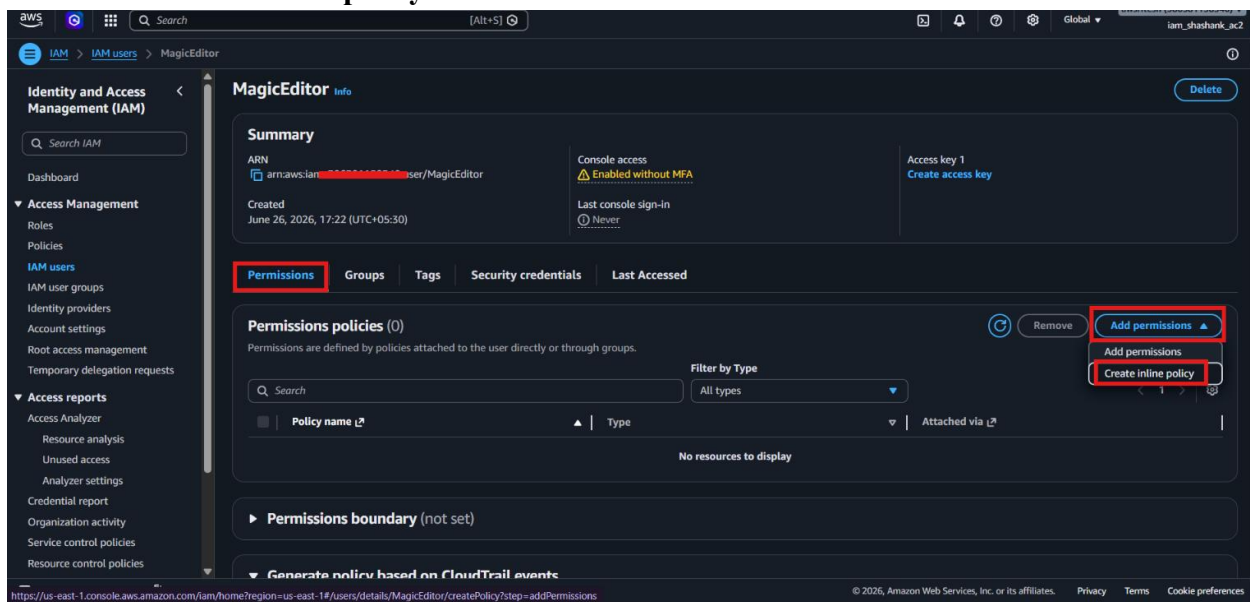


- Leave the user without direct permissions for now.
- Create the user.

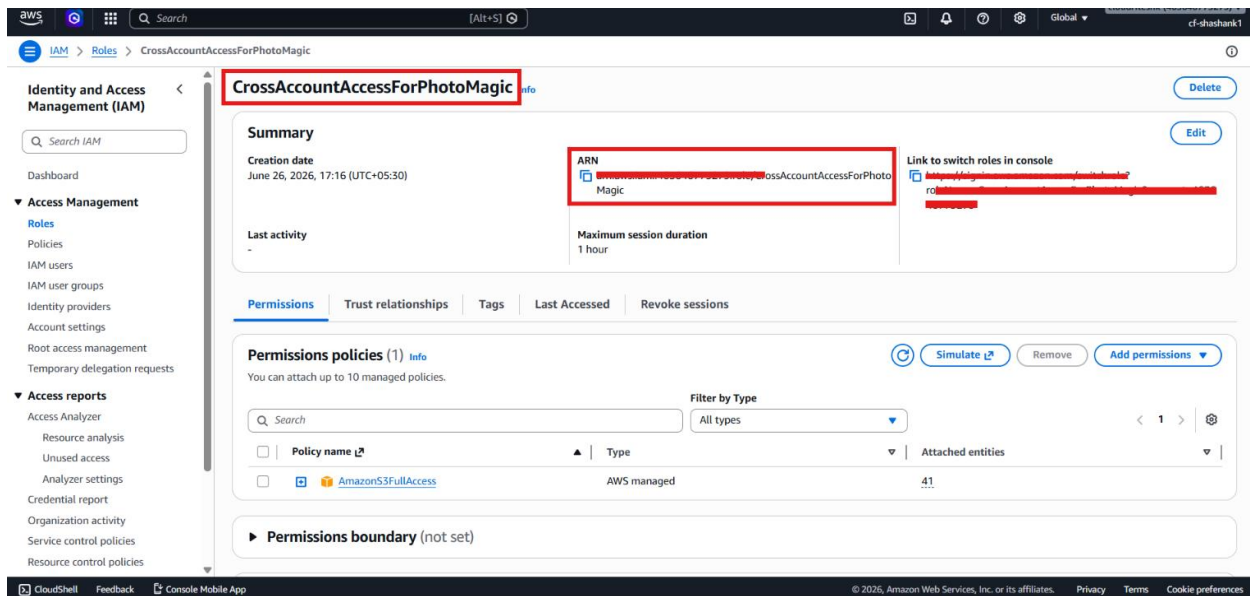


Step 4 – Allow the User to Assume the Cross-Account Role

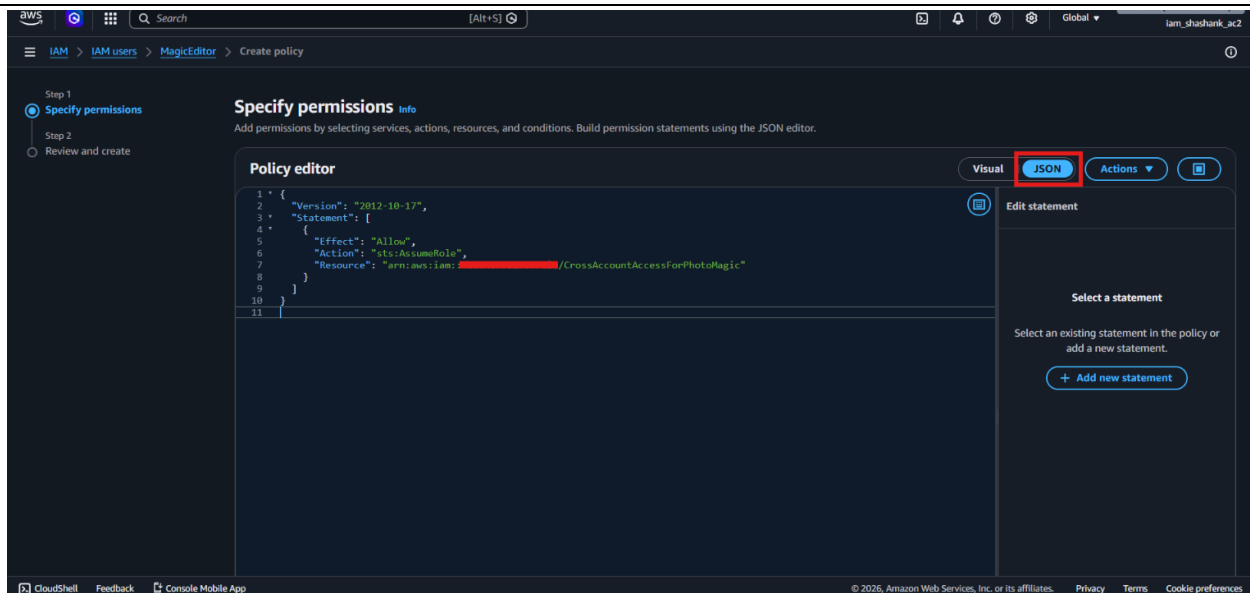
- Open the **MagicEditor** user.
- Go to **Permissions** → **Add permission**.
- Choose **Create inline policy**.



- Open the **JSON** tab.
- Add the policy that allows `sts:AssumeRole` for the Cloud Store role ARN.



```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": "sts:AssumeRole",
      "Resource": "arn:aws:iam::<cloud-store-account>:role/CrossAccountAccessForPhotoMagic"
    }
  ]
}
```



- Name the policy and create the policy.

Step 1: Specify permissions
Step 2: Review and create

Review and create

Review the permissions, specify details, and tags.

Policy details

Policy name
Enter a meaningful name to identify this policy.

CrossAccessPolicy

Maximum 128 characters. Use alphanumeric and '*' '_' '@' '-' characters.

Permissions defined in this policy

Permissions defined in this policy document specify which actions are allowed or denied. To define permissions for an IAM identity (user, user group, or role), attach a policy to it.

Search

Allow (1 of 471 services) Show remaining 470 services

Service	Access level	Resource	Request condition
STS	Limited: Write	RoleName string like [CrossAccountAccessForPhotoMa gic	None

Cancel Previous **Create policy**

Step 5 – Sign in as the Photo Magic User

- Open the IAM user sign-in URL for the Photo Magic account.
- Log in as **MagicEditor**.

Provide feedback Multi-session disabled English

aws

IAM user sign in

Account ID or alias (Don't have?)

Remember this account

IAM username
MagicEditor

Password

Show Password Having trouble?

Sign in

Sign in using root user email

Create a new AWS account

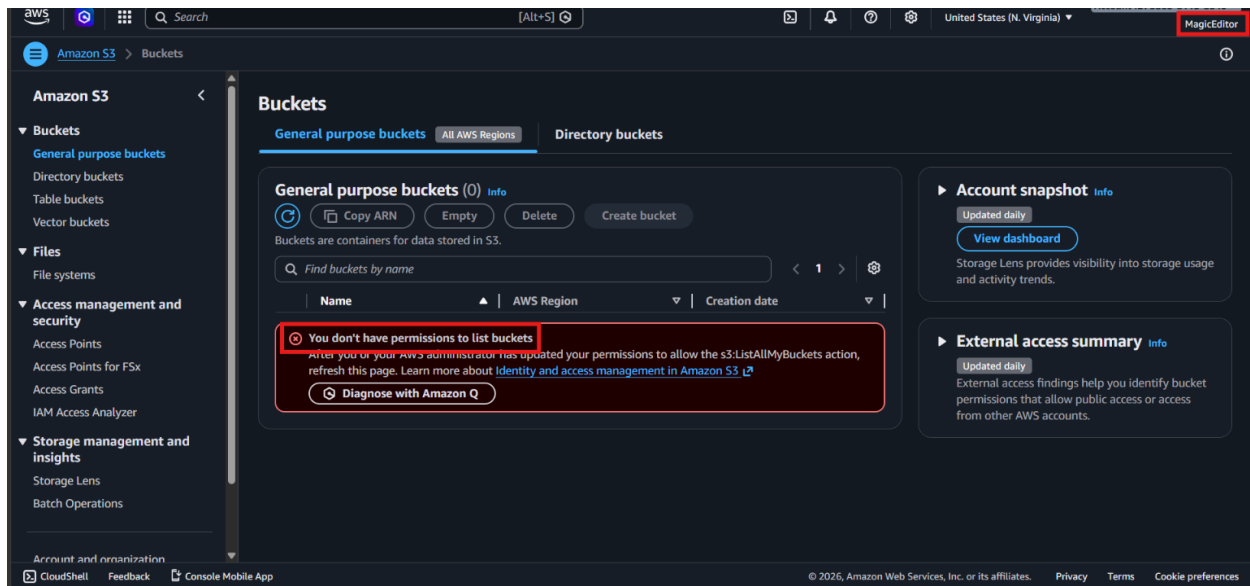
By continuing, you agree to AWS Customer Agreement or other agreement for AWS services, and the Privacy Notice. This site uses essential cookies. See our Cookie Notice for

Grow faster with AI

AI solutions built for small businesses — ready to deploy with trusted AWS Partners

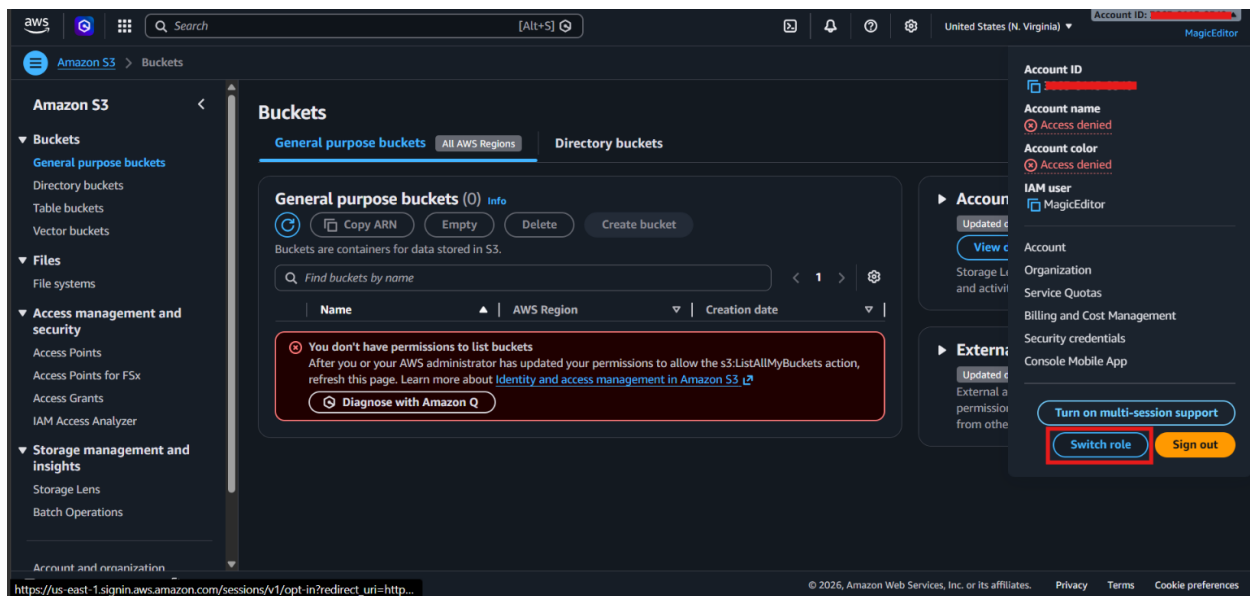
Learn more →

- Verify that the user currently has no direct access to the CloudStore S3 bucket.

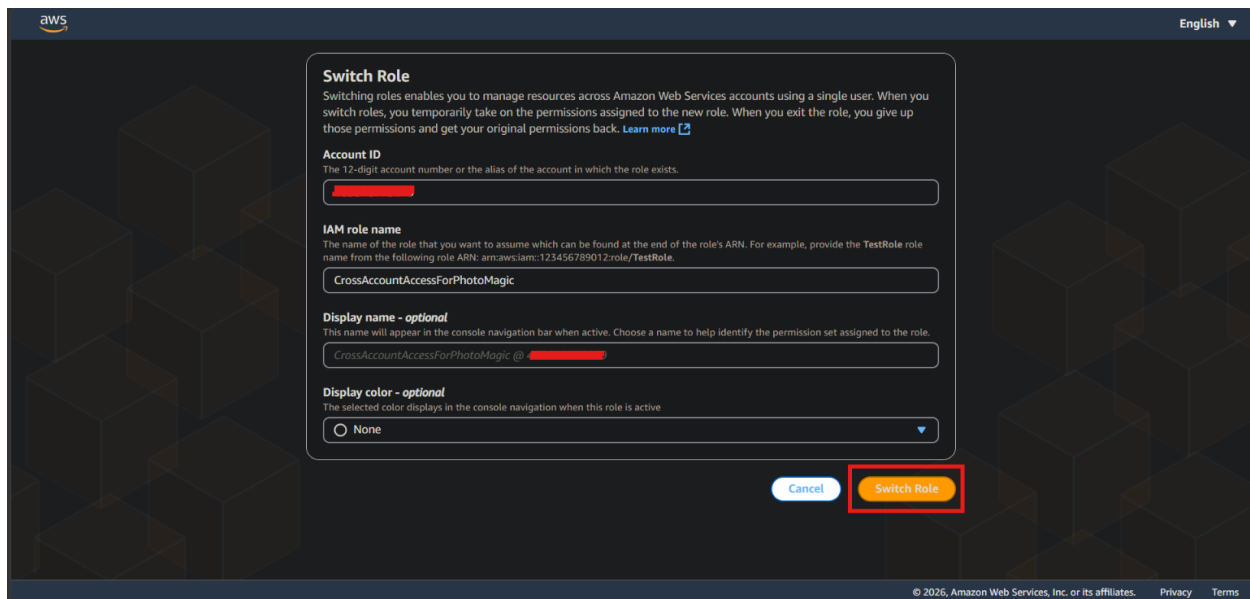


Step 6 – Switch Role to Cloud Store Access

- Open the account menu.
- Click **Switch role**.



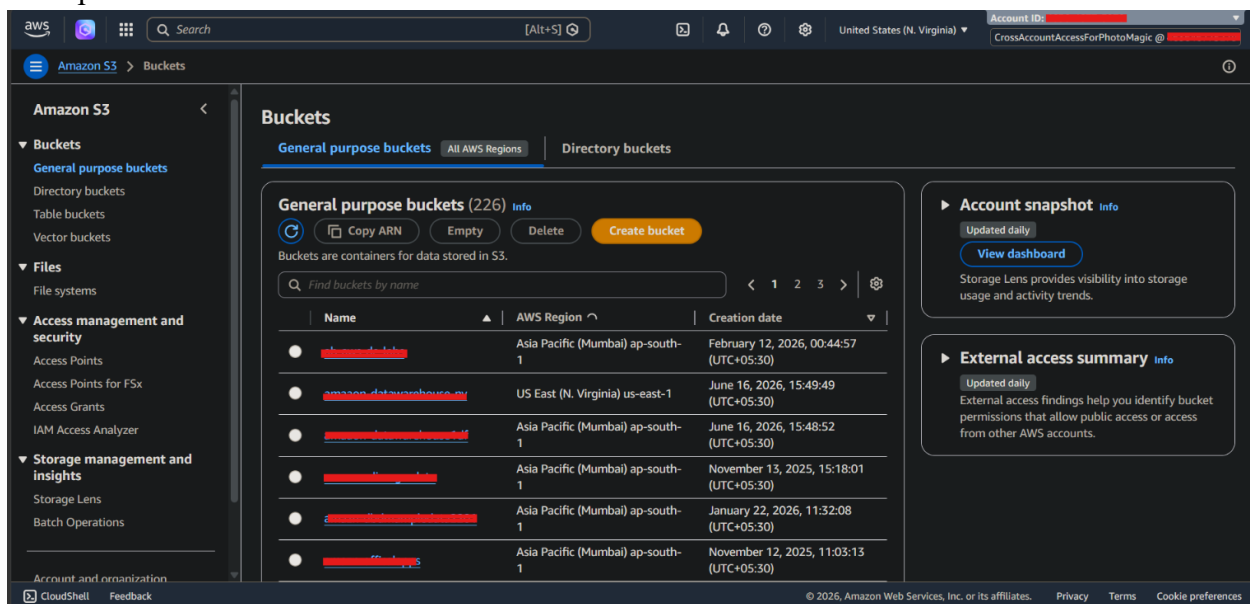
- Enter the Cloud Store AWS account ID.
- Enter the role name **CrossAccountAccessForPhotoMagic**.
- Enter a display name if needed.
- Click **Switch role**.



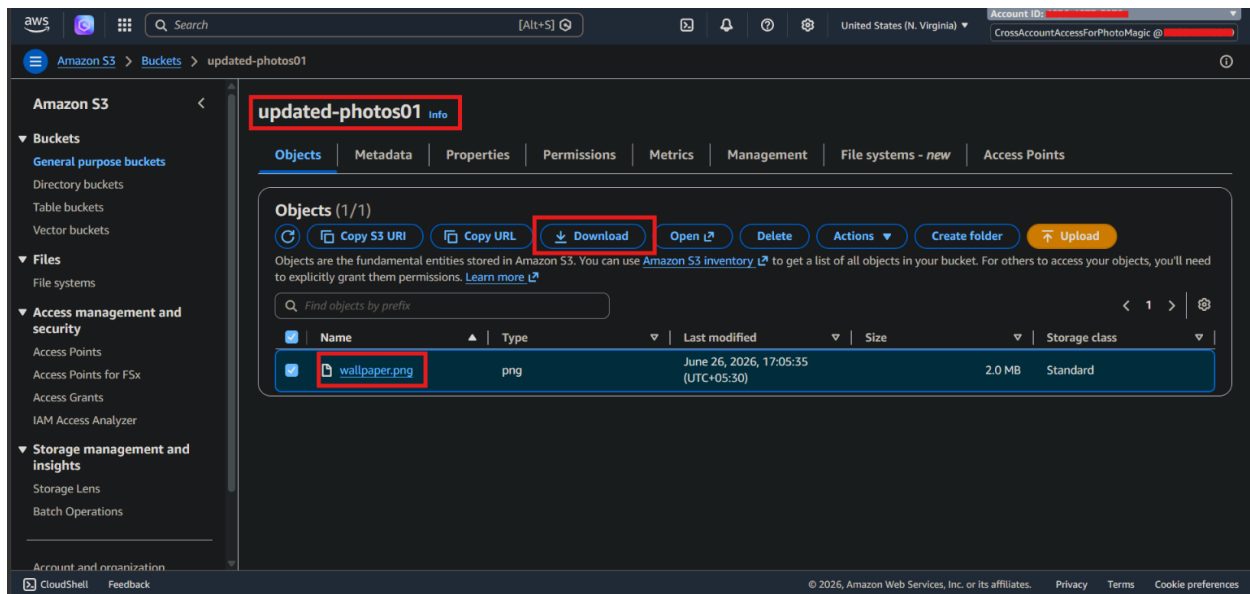
- Confirm that the console now reflects the assumed role session.

Step 7 – Access the Cloud Store S3 Bucket

- Open Amazon S3 while the role is assumed.



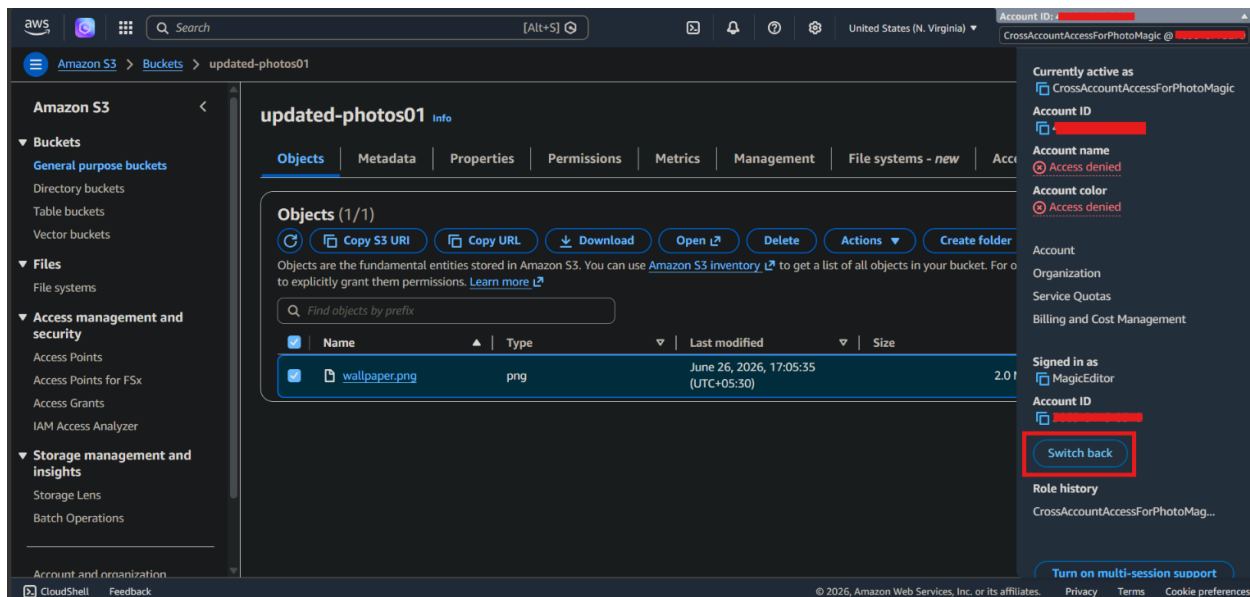
- Select the **updated-pictures01** bucket.
- Verify that the images are visible.
- Download an object from the bucket.



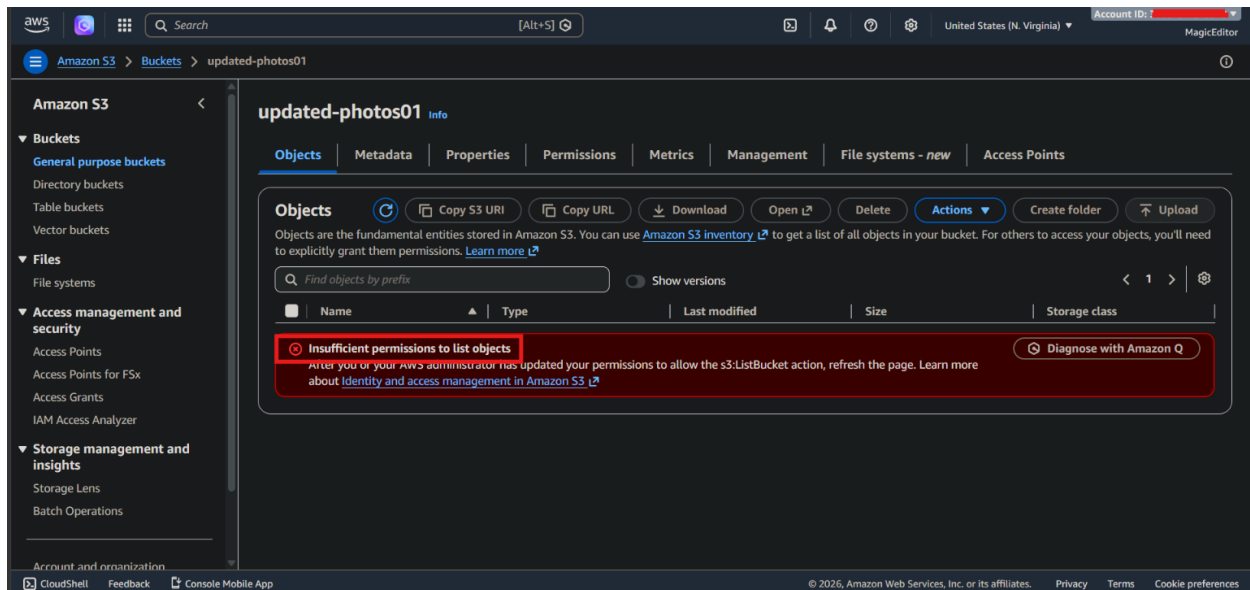
- Upload a test object if required.
- Confirm that the cross-account role provides temporary access only.

Step 8 – Switch Back to the Original Identity

- Open the account menu again.
- Click **Switch back**.



- Return to the normal **MagicEditor** identity.
- Verify that access to the CloudStore S3 bucket is removed again.



Verification

- The Cloud Store S3 bucket was created successfully.
- The cross-account IAM role was created in Cloud Store.
- The Photo Magic IAM user was created successfully.
- The user was allowed to assume the Cloud Store role.
- Temporary access to the S3 bucket worked through **Switch role**.
- The user could download and upload bucket objects while the role was active.
- Switching back removed the temporary access.
- The lab demonstrated secure cross-account access using **AWS IAM assume role**.